

COSC 594: Interplay Between RL and GenAI

Homework 4

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This homework is due on **11/21, 11:59pm**. Write your solution in Latex and submit it on Canvas.

1. (DM for Offline RL)

- (a) We consider a policy $\pi(a^0 | s)$ (a conditional distribution over action a^0 given state s) as a diffusion policy as follows. A diffusion policy consists of two processes: forward process and backward process. The forward process adds the Gaussian noise to the action a^0 step by step by following the Gaussian transition kernel,

$$q(a^t | a^{t-1}, s) = \mathcal{N}(a^t; \sqrt{\alpha_t} a^{t-1}, (1 - \alpha_t)I)$$

where α_t is the noise schedule parameter. This leads to a ground-truth backward process,

$$q(a^{t-1} | a^t, a^0, s) = \mathcal{N}(a^{t-1}; \mu(a^t, a^0, s), \sigma^2(t)I) \quad (1)$$

where the ground-truth mean $\mu(a^t, a^0, s)$ is given by

$$\mu(a^t, a^0, s) = \frac{1}{\sqrt{\alpha_t}} a^t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t} \sqrt{\alpha_t}} \epsilon_0 \text{ for } \epsilon_0 \sim \mathcal{N}(0, I)$$

where $\bar{\alpha}_t := \prod_{i \leq t} \alpha_i$, and the variance is given by $\sigma^2(t) := \frac{(1 - \alpha_t)(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t}$. The backward process reverses the forward process by approximating the ground-truth backward process. It begins with a^T sampled from the standard Gaussian $p(a^T) = \mathcal{N}(a^T; 0, I)$, and iteratively transitions from a^T to a^0 through conditional distributions $p_t^\theta(a^{t-1} | a^t, s)$ for $t = T, \dots, 1$, where θ is the model parameter. This gives a way to sample from $\pi(a^0 | s)$. As we did for the diffusion model, a diffusion policy is represented as a Gaussian with mean $\mu_\theta(a^t, t, s)$ and variance $\sigma^2(t)$,

$$p_t^\theta(a^{t-1} | a^t, s) := \mathcal{N}(a^{t-1}; \mu_\theta(a^t, t, s), \sigma^2(t)I). \quad (2)$$

In particular, we consider a class of denoising diffusion models,

$$\mu_\theta(x^t, t, s) = \frac{1}{\sqrt{\alpha_t}} a^t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t} \sqrt{\alpha_t}} \epsilon_\theta(a^t, t, s)$$

where $\epsilon_\theta(a^t, t, s)$ is the noise predictor that approximates the Gaussian noise.

- i. (1pt) Compute the following KL divergences between the ground-truth backward process (1) and the forward process (2),

$$D_{\text{KL}}(q(a^{t-1} | a^t, a^0, s) \| p_t^\theta(a^{t-1} | a^t, s)).$$

Solution:

- ii. (1pt) We next apply the diffusion policy to the following constrained policy optimization problem,

$$\begin{aligned} & \underset{\pi}{\text{maximize}} && V_r^\pi(\rho) \\ & \text{subject to} && V_g^\pi(\rho) \geq b. \end{aligned}$$

In the offline setting, we only access an offline data set $\mathcal{D} = \{(s_t, a_t, s_{t+1})\}$ from a behavior policy π_β .

Reformulate the constrained policy optimization problem as an offline constrained policy optimization problem. *Hint:* apply performance difference lemma to the value functions $V_r^\pi(\rho)$ and $V_g^\pi(\rho)$, and then approximate the state distribution d_ρ^π by $d_\rho^{\pi_\beta}$.

Solution:

- iii. (1pt) To reduce the mismatch between d_ρ^π and $d_\rho^{\pi_\beta}$, we introduce a KL-divergence regularization $\mathbb{E}_{s \sim d_\rho^{\pi_\beta}} [D_{\text{KL}}(\pi(\cdot | s) \| \pi_\beta(\cdot | s))]$ to the offline constrained policy optimization problem in (ii). Formulate the KL-divergence regularized constrained policy optimization problem and find its closed-form solution. *Hint:* note that strong duality holds for any feasible convex optimization with linear constraint, thus there exists an optimal dual variable, denoted by λ^* , and you can fix this optimal dual variable to find a maximizer of the Lagrangian.

Solution:

- iv. (2pt) Assume that any value functions of the behavior policy π_β are exactly estimated. Design a training algorithm to find an approximate solution of the KL-divergence regularized constrained policy

optimization problem. *Hint*: first replace the policy π by the diffusion policy π_θ ; second, replace the point-wise KL by a path-wise KL and apply the result in (i); finally, design a training algorithm to solve the Lagrangian maximization problem by assuming an optimal dual variable λ^* .

Solution:

2. (DM for Online RL)

- (a) We continue to apply the diffusion policy to the following constrained policy optimization problem,

$$\begin{aligned} \underset{\pi}{\text{maximize}} \quad & V_r^\pi(\rho) \\ \text{subject to} \quad & V_g^\pi(\rho) \geq b. \end{aligned}$$

Now we want to develop an online RL algorithm. A simple algorithm in literature is given by the following primal-dual algorithm,

$$\begin{aligned} \pi^{(t+1)}(\cdot | s) &= \underset{\pi'}{\text{argmax}} \langle Q_r^{\pi^{(t)}}(s, \cdot) + \lambda^{(t)} Q_g^{\pi^{(t)}}(s, \cdot), \pi'(\cdot | s) - \pi^{(t)}(\cdot | s) \rangle - \frac{1}{\eta} D_{\text{KL}}(\pi'(\cdot | s) \| \pi^{(t)}(\cdot | s)) \\ \lambda^{(t+1)} &= \left[\lambda^{(t)} - \eta (V_g^{\pi^{(t)}}(\rho) - b) \right]_+. \end{aligned}$$

We next turn this algorithm to an online RL algorithm using diffusion policy.

- i. (1pt) Find a closed-form expression of the dual function $\pi^{(t+1)}$, and formulate the policy update as a KL-divergence minimization problem.

Solution:

- ii. (1pt) Write the KL-divergence minimization problem in (i) as an exponential Q -weighted likelihood maximization problem.

Solution:

- iii. (1pts) Consider the diffusion policy in 1 (a), write the exponential Q -weighted likelihood maximization problem in (ii) as a path weighted KL-divergence minimization problem.

Solution:

- iv. (2pts) Assume that any value functions are exactly estimated. Rewrite the primal-dual algorithm as a diffusion policy based primal-dual algorithm. *Hint*: evaluate the path weighted KL-divergence in

terms of the diffusion policy in 1 (a), and then replace the policy update by a step of updating diffusion model.

Solution: